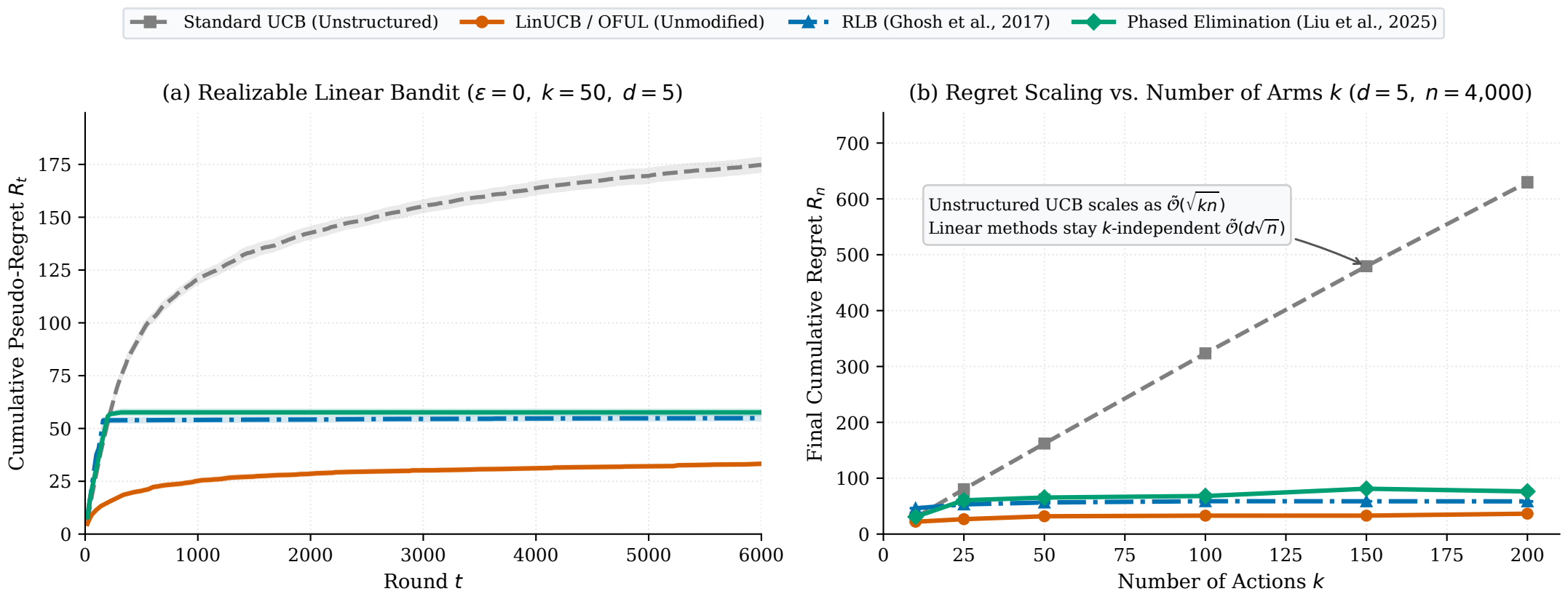
Standard UCB (Unstructured)
LinUCB / OFUL (Unmodified)
RLB (Ghosh et al., 2017)
Phased Elimination (Liu et al., 2025)
(a) Realizable Linear Bandit ($\varepsilon = 0,\ k = 50,\ d = 5$)
Cumulative Pseudo-Regret R_t
Round t
0
25
50
75
100
125
150
175
0
1000
2000
3000
4000
5000
6000
(b) Regret Scaling vs. Number of Arms k ($d = 5,\ n = 4{,}000$)
Final Cumulative Regret R_n
Number of Actions k
Unstructured UCB scales as $\tilde{\mathcal{O}}(\sqrt{kn})$
Linear methods stay k-independent $\tilde{\mathcal{O}}(d\sqrt{n})$
0
100
200
300
400
500
600
700
0
25
50
75
100
125
150
175
200